

Carbon Leakage to Developing Countries

Diego Känzig

Northwestern, CEPR & NBER

Julian Marenz

London Business School

Marcel Olbert

London Business School

BREAD/Development Economics Program Meeting, Fall 2025

Outline

Motivation

Measuring leakage

Empirical design & micro evidence

Macro evidence

Conclusion

Fighting climate change through regulation

- Looming **climate crisis** put climate change at top of global policy agenda
- **Carbon pricing** seen as key tool to **mitigate** climate change, but:
 - Very difficult to implement globally
 - More progress at the national / regional level
- National policies create incentives to shift emissions abroad $\Rightarrow$ **carbon leakage**
 - **Developing countries** attractive destinations because of laxer regulation
 - Undermines goal of net zero and raises distributional questions

This paper

- We study **carbon leakage** from developed to developing countries
 - Focus on how climate policies in Europe affect emissions in Africa
 - Africa prime destination for leakage given proximity and lax environmental regulations
- Propose **new approach** to measure carbon leakage to developing countries
 - Leverages **locations** of multinationals' subsidiaries and **spatial emissions data**
- Exploit changes in regulatory exposure to identify causal leakage effects
 - Within multinationals, at country-level, and in the aggregate

Preview of results

- We find significant **carbon leakage** within multinational firms
 - Emissions at African subsidiaries of regulated multinationals increases
 - Positive employment effects in vicinity of subsidiaries
- Emissions in Europe fall but so does investment & employment
- Evidence for carbon leakage beyond multinational networks
 - Comparable relative effects at country-level
 - Based on aggregate data, we document a leakage rate of $\approx 20\%$

Related literature

- Evidence on carbon leakage is **sparse**
 - Climate policies not binding enough & leakage difficult to accurately measure
 - **Existing approaches:**
 - Carbon content of bilateral trade flows
Aichele and Felbermayr, 2015, Naegele and Zaklan, 2019
 - Survey data on carbon emissions of multinational firms
Ben-David et al., 2021, Dechezleprêtre et al., 2022
 - Administrative data
Cui et al., 2022, Chen et al., 2023, Colmer et al., 2025
- ▷ Results are inconclusive and measurement premature
- We contribute by providing new evidence on leakage effects to developing countries

Outline

Motivation

Measuring leakage

Empirical design & micro evidence

Macro evidence

Conclusion

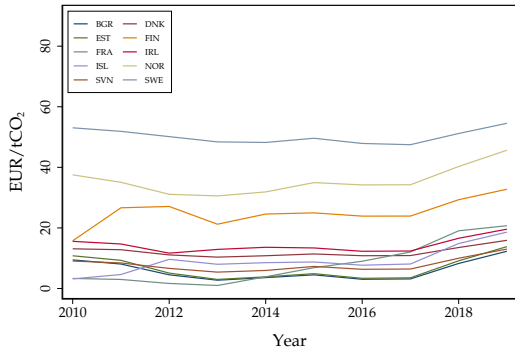
Policy setting and identifying variation

- **Europe** as global leader in climate policy. Two main policy pillars:
 1. Carbon market: Emissions Trading System (ETS)
 2. National carbon taxes
- We exploit effective carbon price: **Countries' carbon taxes + ETS prices**
 - ⇒ Multinationals' **exposure** to these taxes varies greatly
 - **Idea**: compare more regulated to less regulated multinationals

Effective carbon prices

Stringent countries

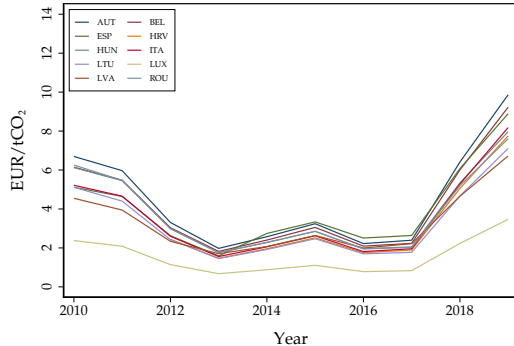
Effective carbon prices



► More Details

Laxer countries

Effective carbon prices



Carbon leakage within multinational firms

- Identify global subsidiary networks of MNE's using Orbis [▶ Orbis Detail](#)

As in Hoopes, Klein, Lester and Olbert 2023, Copolla, Maggiori, Neiman, Schreger 2021

- Focus on subsidiaries active in Africa and owned by UP with operations in Europe

 $\approx 1,500$ multinationals from around the world with $>15,000$ subsidiaries in Africa

 Manufacturing + heavy industries + ETS covered firms

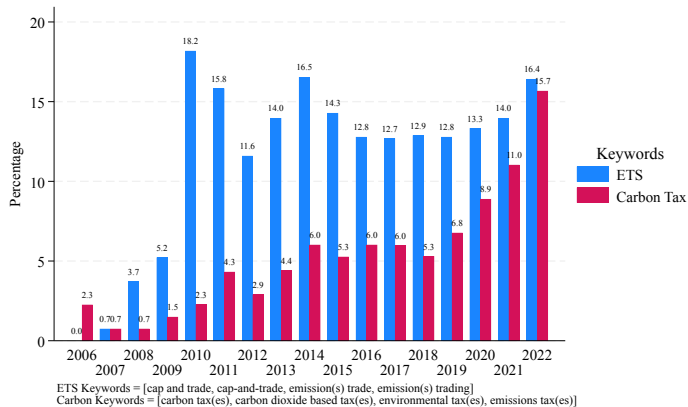
 Link MNE (parent) exposed to EU regulation and location of operations in Africa

- Challenges:

- Climate policy binding enough?
- How to measure emissions at African subsidiaries?

Climate policy is financially material

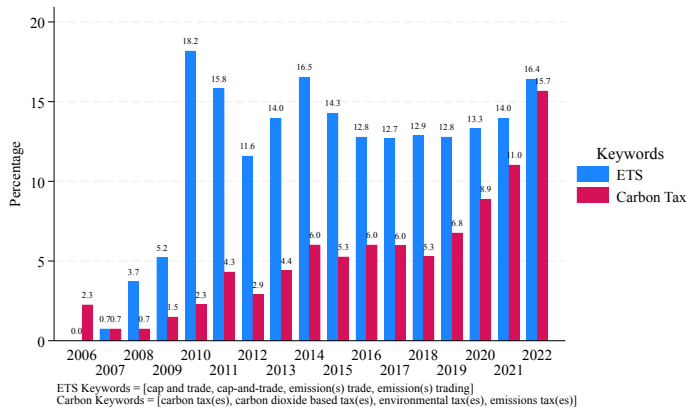
Evidence from sample multinationals' annual reports:



► [More](#)

Climate policy is financially material

Evidence from sample multinationals' annual reports:



- Carbon prices increasingly discussed in annual reports
- Financially material regulatory risk

► More

Emissions at African subsidiaries

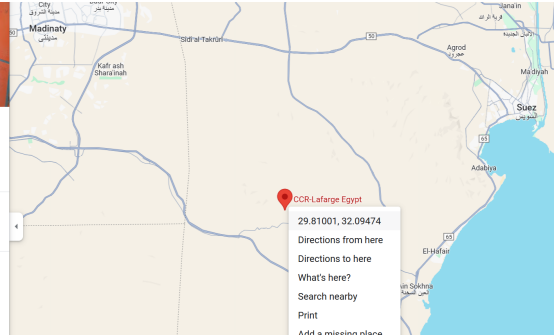
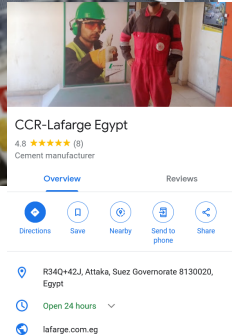
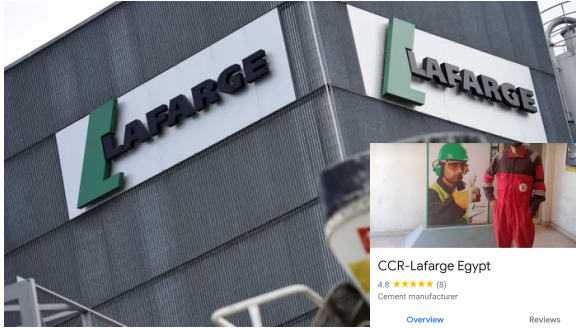
- Financial info, let alone emissions data largely unavailable for African subsidiaries
- **Our approach:**
 1. Georeference the subsidiary locations in Africa
⇒ hand collection
 2. Proxy emissions using spatial emissions data around subsidiary locations
⇒ EDGAR

Georeferencing of African subsidiaries

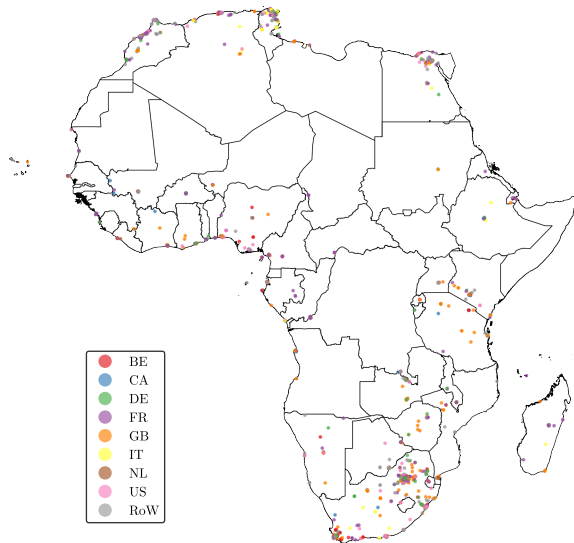
- We georeference African subsidiaries in a large hand-collection effort

Georeferencing of African subsidiaries

- We georeference African subsidiaries in a large hand-collection effort



Georeferencing of African subsidiaries



- **Lafarge:** 43 African subsidiaries in 12 different African countries
- **General Electric:** 16 African subsidiaries in 7 different African countries

Greenhouse gas emissions in Africa from EDGAR

- Emission Database for Global Atmospheric Research (**EDGAR**)

Maintained by the European Commission Joint Research Centre and the Netherlands Environmental Assessment Agency

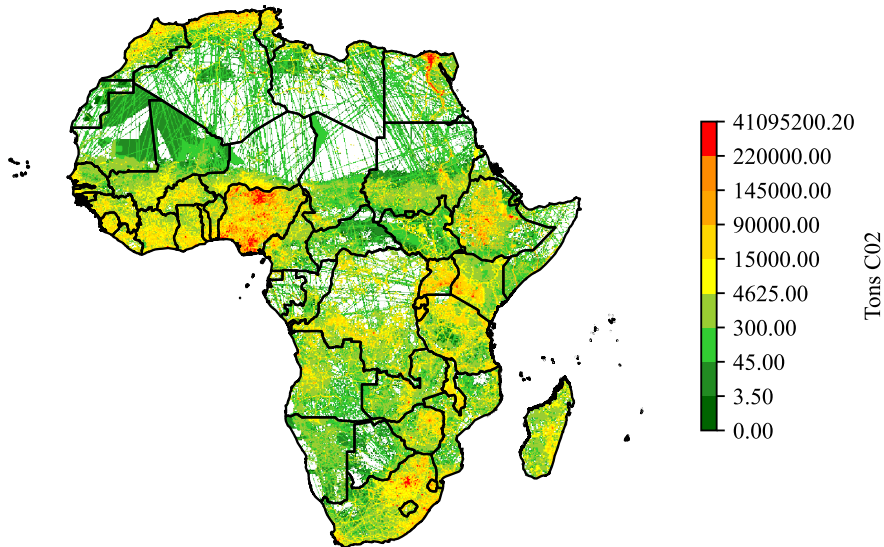
- Resolution 0.1×0.1 degree (c. $11\text{km} \times 11\text{km}$ at equator)

- Emissions are distributed from standard annual statistics on fuel, products, waste, crops, and livestock at the country level using spatial proxy datasets ▶ [EDGAR details](#)

- We validate with spatial luminosity data and aggregate emissions from World bank
▶ [EDGAR validation](#)

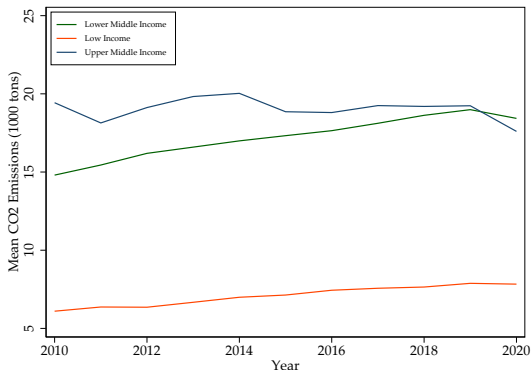
Our outcome of interest: Emissions in close proximity around subsidiary locations

Greenhouse gas emissions in Africa: EDGAR coverage

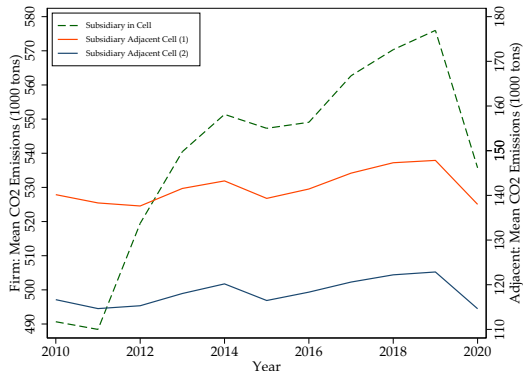


Greenhouse gas emissions in Africa: EDGAR grid-cell variation

(a) AVERAGE BY COUNTRY INCOME GROUP



(b) MNE SUBSIDIARY AND ADJACENT



Outline

Motivation

Measuring leakage

Empirical design & micro evidence

Macro evidence

Conclusion

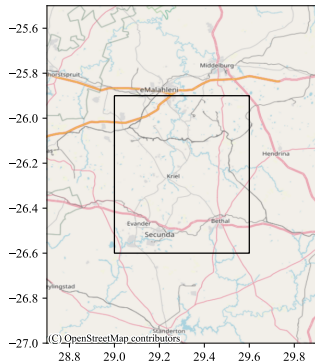
An illustrative example

- France introduced a carbon tax in 2014, Germany has no such tax
- If multinationals offshore pollution, we should see emissions increase at subsidiaries **affected** by the tax vis-à-vis subsidiaries that are **not**

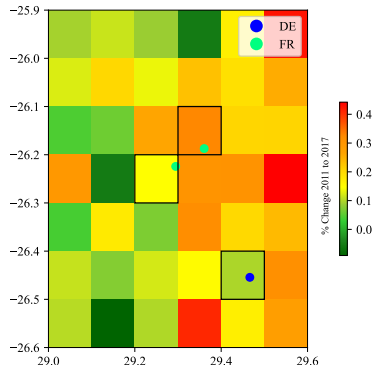
An illustrative example

- France introduced a carbon tax in 2014, Germany has no such tax
- If multinationals offshore pollution, we should see emissions increase at subsidiaries **affected** by the tax vis-à-vis subsidiaries that are **not**

(a) Zoom area



(b) CO2 emissions



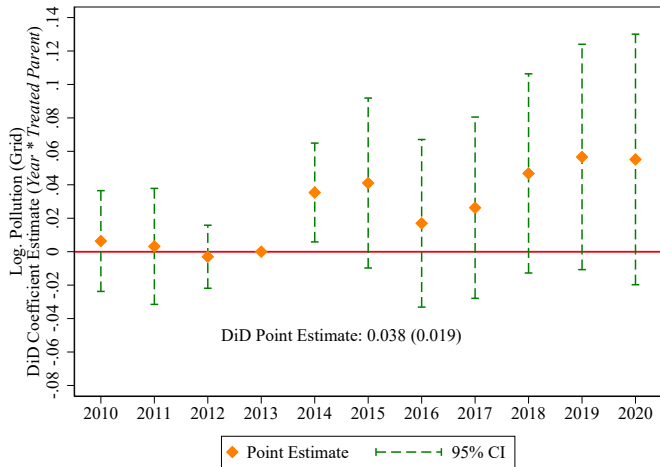
An event study of French carbon tax

- Does this hold up more systematically?
- Event study / diff-in-diff:

$$\ln(\text{CO2}_{g(c)t}) = \alpha_{g(c)} + \kappa_c \times \delta_t + \sum_{r \neq -1} \beta_r \times \mathbf{1}[\tau_t = r] \times \text{Treat}(S)_{g(c)} + \epsilon_{g(c)t}$$

- Proxy exposure by home country of parent
 - **Treated:** Subsidiaries of French multinational firms subject to carbon tax since 2014
 - **Control:** Subsidiaries of German multinationals
 - Include only firms with significant exposure to home-jurisdiction

An event study of French carbon tax



- Significant carbon leakage, but France only one example ...

Measuring the full exposure to European carbon policy

- Firms may have significant carbon price **exposure** in countries other than at home
 - Construct **exposure measure** based on
 - Country-level effective carbon prices (ETS prices + national taxes)
 - Share of operations in given country
- leverage unique subsidiary-level financial statement data in Europe

Measuring the full exposure to European carbon policy

- Firms may have significant carbon price **exposure** in countries other than at home
 - Construct **exposure measure** based on
 - Country-level effective carbon prices (ETS prices + national taxes)
 - Share of operations in given country
- leverage unique subsidiary-level financial statement data in Europe

$$\text{CarbonPolicyExp}_{it} = \sum_{d \in D} w_{id} \times \text{cprice}_{dt} \quad \text{with} \quad w_{id} = \left(\sum_{t=2007}^{2009} \frac{\text{Assets}_{idt}}{\sum_{k \in K} \text{Assets}_{ikt}} \right) / 3$$

- Assets_{idt} are unconsolidated fixed tangible assets
- Captures also indirect carbon price exposure via suppliers (e.g. higher energy prices)

Validating the carbon policy exposure measure

	(1)	(2)	(3)	(4)	(5)
	CO2 Policy Words/1k Words	CO2 Policy Risk Disclosure (%)			
Carbon Policy Exp. (FTanA)	0.98* (0.5356)	4.87* (2.4668)			
Carbon Policy Exp. (FTanA) _{t-1}			6.13** (2.9420)		
Carbon Policy Exp. (FTanA) _{t-2}				13.58*** (5.1534)	
Carbon Policy Exp. (FTanA) _{t-3}					12.35** (4.7530)
Obs.	1,612	1,612	1,502	1,399	1,295
Adj. R2	0.595	0.751	0.800	0.856	0.904
Firm FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes

Our *CarbonPolicyExp* measure predicts **firm-level risk disclosures** re carbon policies!

Exposure design

- Challenge: if subsidiaries are close, our emissions data is not granular enough
- Work at grid-cell level:

$$\ln(\text{CO2}_{gt}) = \alpha_g + \delta_t + \beta \text{CarbonPolicyExp}_{gt-1} + \epsilon_{gt}$$

- $\text{CarbonPolicyExp}_{gt-1}$ is the avg. grid-cell exposure based on subsidiaries in cell g
- **Leverages:**
 - multinationals' pre-determined exposure to European carbon prices
 - time-series variation in countries' carbon prices
- Start with grid-cell and time FE, saturate with more conservative FE

Carbon leakage within multinational firms - grid-cell-level analysis

	(1)	(2)	(3)
Panel A	ln(CO2)		
Carbon Policy Exp. (FTanA)	0.1159*** (0.0309)	0.1003*** (0.0310)	0.1248*** (0.0353)
Obs.	7,227	7,227	7,128
R2	0.021	0.014	0.027
Grid Cell FE	Yes	Yes	Yes
Year FE	Yes		
African Region $\times$ Year FE	No	Yes	
African Country $\times$ Year FE	No	No	Yes

Carbon leakage within multinational firms - grid-cell-level analysis

	(1)	(2)	(3)
Panel A	ln(CO2)		
Carbon Policy Exp. (FTanA)	0.1159*** (0.0309)	0.1003*** (0.0310)	0.1248*** (0.0353)
Obs.	7,227	7,227	7,128
R2	0.021	0.014	0.027
Grid Cell FE	Yes	Yes	Yes
Year FE	Yes		
African Region $\times$ Year FE	No	Yes	
African Country $\times$ Year FE	No	No	Yes

- Higher exposure leads to **significant carbon leakage** to African subsidiaries
- 1 SD (5 USD/tCO₂e in Europe) $\Rightarrow$ 10-13% more local CO₂ emissions

Carbon leakage within multinational firms - dynamics

	(1)	(2)	(3)
	$\Delta \ln(\text{CO}_2)$		
	$t_{+1} - t_{-1}$	$t_{+2} - t_{-1}$	$t_{+3} - t_{-1}$
Δ Carbon Policy Exp. (FTanA)	0.0979*** (0.0369)	0.1217*** (0.0441)	0.1260*** (0.0412)
Obs.	5,184	4,536	3,888
R2	0.012	0.024	0.034
African Country $\times$ Year FE	Yes	Yes	Yes

- Consistent with baseline results

 effects immediate but unfolding gradually

Robustness

- Alternative carbon policy exposure measures (total assets, empl., ...)
- Robust to excluding Covid-19 period
- Robust to excluding South Africa
- Robust to winsorization, ...
- Alternative data construction: subsidiary-level

Heterogeneity by environmental standards

	(1)	(2)	(3)	(4)
Panel A:	ln(CO2)			
Countries' Closeness to Environ. Policy Targets	< median	≥ median	Lowest Quintile	Highest Quintile
Carbon Policy Exp. (FTanA)	0.1810* (0.0945)	0.1090*** (0.0358)	0.2666** (0.1166)	0.0192 (0.0417)
Obs.	2,563	4,664	1,573	880
R2	0.035	0.020	0.070	0.001
Grid Cell FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Panel B:	ln(CO2)			
African Countries' Income Group	Low and Lower Middle	Upper Middle	< median 2010 GDP/pc	≥ median 2010 GDP/pc
Carbon Policy Exp. (FTanA)	0.0943* (0.0492)	0.0637** (0.0319)	0.2284*** (0.0884)	0.0755*** (0.0253)
Obs.	4,587	2,607	3,157	3,971
R2	0.011	0.009	0.066	0.012
Grid Cell FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Taking stock

- **Robust** evidence of carbon leakage from Europe to Africa
- Magnitudes not negligible given relatively low tax rates

? Mechanisms and broader consequences: we examine

1. European operations
2. Local economic effects in Africa
3. Leakage beyond multinationals
4. Leakage at aggregate level

The effect on European operations

- How are European operations affected?
- Estimate similar equation

$$\ln(Y_{s(i)t}) = \alpha_s + \delta_t + \beta \text{CarbonPolicyExp}_{it-1} + \epsilon_{s(i)t},$$

for our sample of **European subsidiaries**

- Study effect on **investment** and **employment**

The effect on European operations

	(1)	(2)	(3)
	All EU 28	CO2 Price ≤ 10	CO2 Price > 10
Panel A	ln(Fixed Tan. Assets)		
Carbon Policy Exp. (FTanA)	-0.1035* (0.0599)	-0.0507 (0.0834)	-0.1181* (0.0716)
Panel B	ln(Employees)		
Carbon Policy Exp. (FTanA)	-0.0480 (0.0292)	-0.0128 (0.0402)	-0.0443 (0.0440)
Obs.	54,379	26,245	28,065
R2	0.937	0.942	0.933
Sub. FE	Yes	Yes	Yes
Firm HQ Country $\times$ Year FE	Yes	Yes	Yes

 **Negative effect** on investment and employment in Europe

- Driven by European subsidiaries in countries with **high carbon prices**

Effects at the local economy level in Africa

- Grid-cell analysis using survey data from DHS
- ...*"accurate and representative data on population, health, HIV, and nutrition through more than 400 surveys in over 90 countries."*
- Geo-coded at the individual survey respondent level
- Used in prior work to study local employment
(Hjort and Poulsen, 2019; Hoopes et al. 2023)

Effects at the local economy level in Africa

- Grid-cell analysis using survey data from DHS

	(1)	(2)	(3)	(4)
Panel A	Employment (0/1)			
Carbon Policy Exp. (FTanA) _{t-1}	0.0061* (0.0035)			
Carbon Policy Exp. (FTanA) _{t-2}		0.0064* (0.0035)		
Carbon Policy Exp. (FTanA) _{t-3}			0.0068* (0.0036)	0.0072** (0.0033)
Obs.	118,968	118,968	118,968	118,968
Country FE	Yes	Yes	Yes	
Year FE	Yes	Yes	Yes	
Country × Year FE	No	No	No	Yes

Effects at the local economy level in Africa

	(1)	(2)	(3)	(4)
Panel B	Anemia (0/1)			
Carbon Policy Exp. (FTanA) _{t-1}	-0.0025 (0.0046)			
Carbon Policy Exp. (FTanA) _{t-2}		-0.0021 (0.0044)		
Carbon Policy Exp. (FTanA) _{t-3}			-0.0016 (0.0045)	-0.0015 (0.0046)
Obs.	32,408	32,408	32,408	32,407
Country FE	Yes	Yes	Yes	
Year FE	Yes	Yes	Yes	
Country × Year FE	No	No	No	Yes

 Some evidence for positive employment effects in local communities

 No evidence for (short and mid-term) health consequences (alt: % children coughing)

Outline

Motivation

Measuring leakage

Empirical design & micro evidence

Macro evidence

Conclusion

Carbon leakage beyond multinational firm networks

- Evidence for leakage within multinationals. What about country-level impacts?

$$\ln(\text{CO2}_{ct}) = \alpha_c + \delta_t + \mathbf{X}_{ct-1}\theta + \beta \text{CarbonPolicyExp}_{ct-1} + \epsilon_{ct}.$$

$$\text{CarbonPolicyExp}_{ct} = \sum_{d \in D} w_{cd} \times \text{cprice}_{dt},$$

- **Exploit:**
 - developing countries differential exposure to European carbon prices via **value of bilateral exports** alt: share of subsidiaries owned by EU parents
 - time-series variation in European countries' carbon prices

Leakage at country-level

	(1)	(2)	(3)	(4)
	ln(CO2 Country)			
Carbon Policy Exp. (Macro, Trade)	0.0332* (0.0184)		0.0504** (0.0237)	
Carbon Policy Exp. (Macro, Firms)		0.0464*** (0.0171)		0.0707*** (0.0188)
Obs.	561	561	508	508
R2	0.997	0.997	0.997	0.997
Country FE	Yes	Yes		
Year FE	Yes	Yes		
Macro controls	No	No	Yes	Yes
African region $\times$ Year FE	No	No	Yes	Yes

- Considerable leakage even outside firm boundaries
- Effect somewhat smaller than within multinationals

▷ Illustrates relevance of leakage through multinationals

Leakage in the aggregate

- Aggregate emissions data for European and African countries
- Exploit time-series variation in carbon prices (Metcalf and Stock, 2023)
- Estimate effect using local projections:

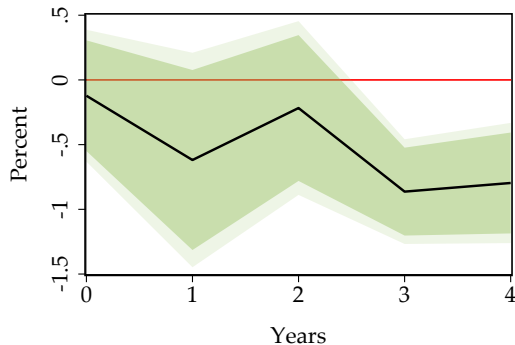
$$\ln(\text{CO2}_{i,t+h}) - \ln(\text{CO2}_{i,t-1}) = \alpha_i + \beta \text{cprice}_{eu,t} + \phi(L) \Delta \ln(\text{CO2}_{i,t-1}) + \delta(L) X_{eu,t-1} + \epsilon_{i,t+h}$$

for $i = \{eu, af\}$

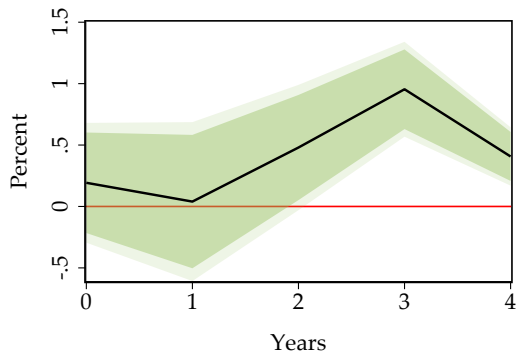
- Start at introduction of EU ETS in 2005

Leakage in the aggregate

(a) European Union



(b) Africa



- Increase in effective carbon price leads to fall in EU emissions
- But emissions in Africa increase
- Estimated leakage rate is 18.5%, but considerable uncertainty: 90% CI: [6.9, 59.2]

Outline

Motivation

Measuring leakage

Empirical design & micro evidence

Macro evidence

Conclusion

Conclusion and Outlook

- Significant **carbon leakage** via multinationals to global south
- Regulating countries adversely affected, positive effects on local growth
- Leakage quantitatively important, $\approx 20\%$ in aggregate
 - Total Energies Climate Report: Scope 1 is 41m worldwide, 21m in EU, 11m in Africa
- Raises important **distributional questions**
- How to design **CBAM** balancing **environmental** and **development** considerations?
 - Currently no exceptions for developing countries
 - Given laxer climate policy stance puts them at a disadvantage

Thank you!

Policy Setting and Identifying Variation

Country	Carbon tax rate	Tax coverage	ETS price	ETS coverage	Effective carbon price
Austria			24.45	40%	9.87
Belgium			24.45	37%	9.23
Bulgaria			24.45	51%	12.29
Croatia			24.45	31%	7.75
Cyprus			24.45	45%	11.20
Czechia			24.45	50%	12.11
Denmark	23.33	40%	24.45	26%	15.90
Estonia	1.94	3%	24.45	57%	13.82
Finland	61.09	36%	24.45	43%	32.77
France	44.04	35%	24.45	21%	20.73
Germany			24.45	44%	10.81
Greece			24.45	46%	11.54
Hungary			24.45	33%	7.97
Iceland	29.45	29%	24.45	42%	18.61
Ireland	19.19	49%	24.45	42%	19.59
Italy			24.45	33%	8.18
Latvia	4.39	15%	24.45	25%	6.72
Lithuania			24.45	29%	7.11
Luxembourg			24.45	14%	3.47
Malta			24.45	41%	10.19
Netherlands			24.45	44%	10.86
Norway	53.35	62%	24.45	50%	45.65
Poland	0.07	4%	24.45	46%	11.31
Portugal	12.52	29%	24.45	34%	11.95
Romania			24.45	32%	7.61
Slovakia			24.45	49%	12.02
Slovenia	16.93	24%	24.45	36%	13.01
Spain	14.81	3%	24.45	34%	8.89
Sweden	112.09	40%	24.45	40%	54.54
United Kingdom	19.93	23%	24.45	26%	11.09

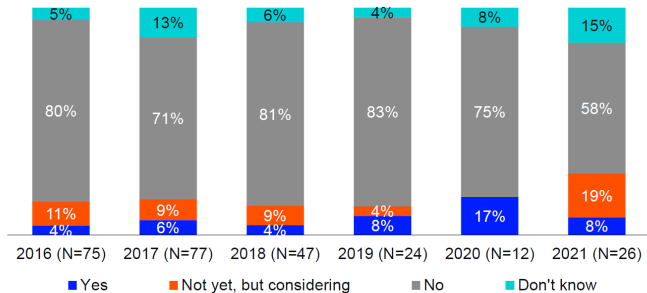
ORBIS Details

- The ownership tree is constructed in the following way:
 1. Identify the ultimate parent (UP): A firm that either publishes consolidated financial statements or that has no owner with $> 50\%$
 2. Identify all firms that are owned by an ultimate owner from 1. with $> 50\%$
 3. Identify all firms that are owned by a firm from 2. with $> 50\%$, or that is owned with $> 50\%$ by direct parents that have the same UP
 4. Repeat until all subsidiaries are identified
- We identify relevant GUO companies as having NAICS code 21 (mining, quarrying, and oil and gas extraction) or 31-33 (manufacturing). Moreover, we include JRC-EU ETS-Firms [▶ Back to Presentation](#)

Carbon Leakage – Likely to happen within (some) firms

Figure 1.9. Location effect

"Has your company moved operations/investments outside the EU ETS because of carbon costs?"*



* Until 2019 this question read: "Has your company moved production/activity outside the EU ETS because of carbon costs?"

Source: Refinitiv

Consistent with [Martin et al. \(AER, 2014\)](#) data from telephone interviews.

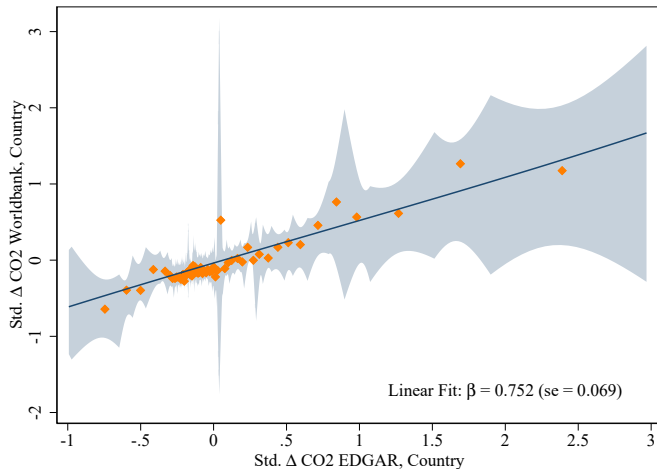
EDGAR Details

A geographical database was built using spatial proxy datasets with the location of energy and manufacturing facilities, road networks, shipping routes, human and animal population density and agricultural land use, that vary over time. The input datasets where point, line and area grids at various resolutions and using GIS techniques for conversion, resampling and aggregation the dataset have been included on a $0.1^\circ \times 0.1^\circ$ grid. National sector totals are then distributed with the given percentages of the spatial proxies over the country's area.

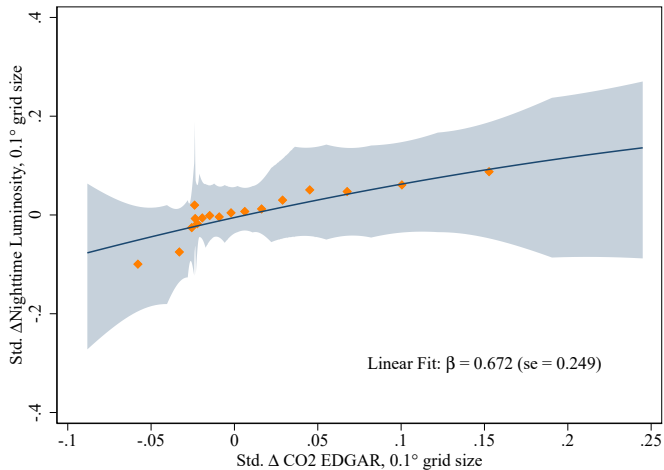
Additional insights from [EDGAR v8.0 documentation](#): 130 proxy datasets, including satellite data, serve as proxy datasets. Examples of time-series information at the local level: for power plants, the dataset tracks the location and also have information on capacity. For venting and flaring as well as intensive livestock and fertilizer-manufacturing industries, the dataset also captures the intensive margin changes over time they track these activities using satellite/spaceborne measurements.

[▶ Back to Presentation](#)

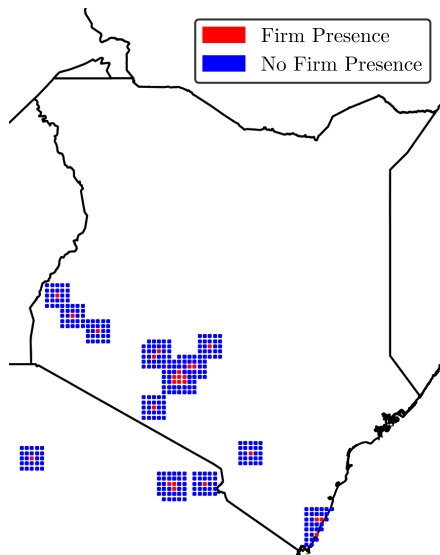
EDGAR Validation: Country-level Changes



EDGAR Validation: Grid-Cell Level Changes

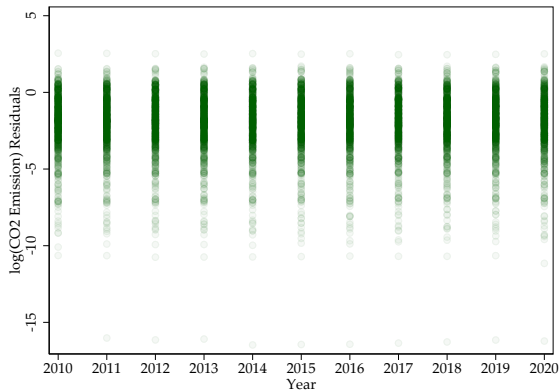


EDGAR Validation: Selecting Grid Cells



EDGAR Validation: Residual variation

(a) COUNTRY $\times$ YEAR



(b) COUNTRY $\times$ YEAR AND CELL FE

